Designing Logic

Door closed AND start pressed: then it fills.



National 5 Engineering Science

Daily Review



1	Z = A + B. **Name** the gate.	OR
2	**Write** the Boolean for **NOT A**.	$\overline{A}$
3	**State** how many rows for **3** inputs.	8
4	**State** another name for a **NOT** gate.	Inverter
5	**State** the **unit** of **current**.	Ampere (A)

PART 1

The Design Method

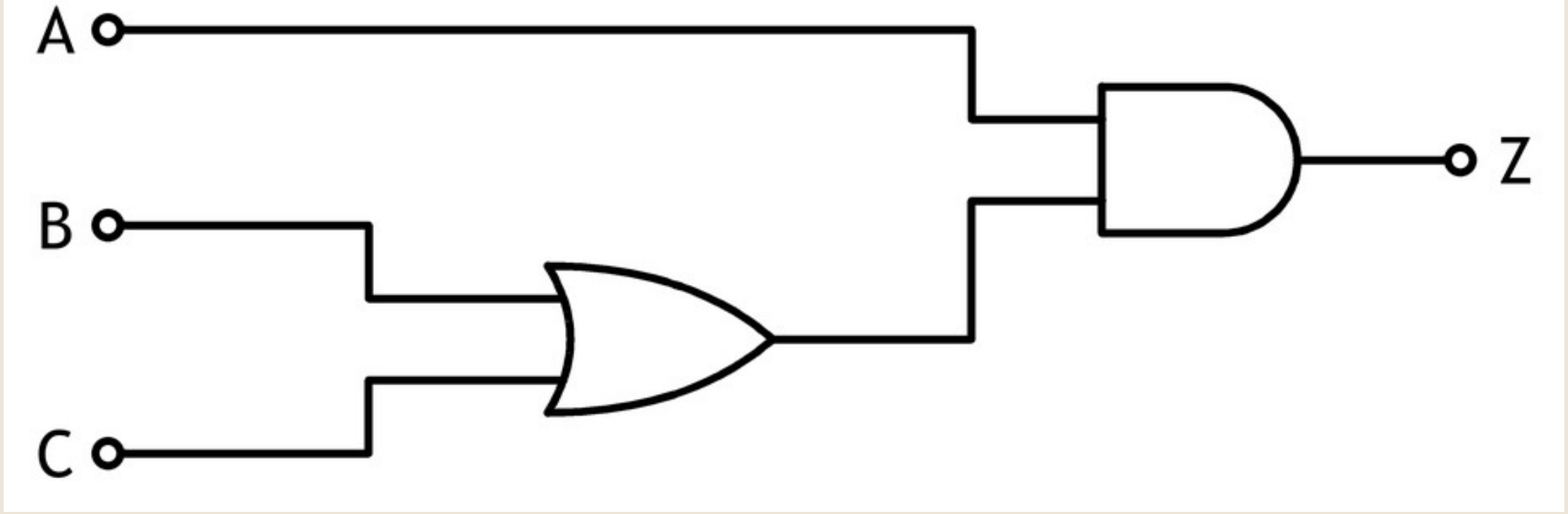
We are learning to design a logic circuit for a real control problem.

By the end of this part…



1. I can **pick out** the **inputs** and the **output**
2. I can **choose** AND, OR or NOT and **write** the Boolean
3. I can **draw** the diagram and **check** it with a truth table

A lift moves when the door is closed AND a button is pressed. Which gates?



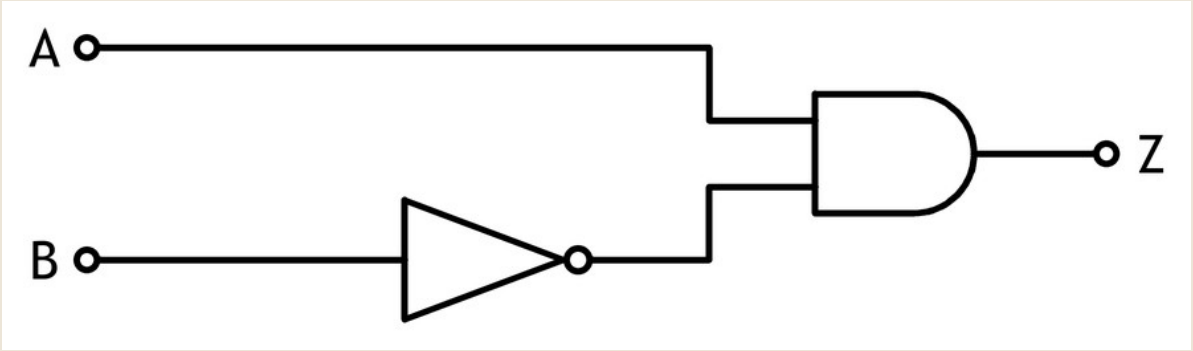
Four Steps to a Design

1 **Underline** the inputs and the output.

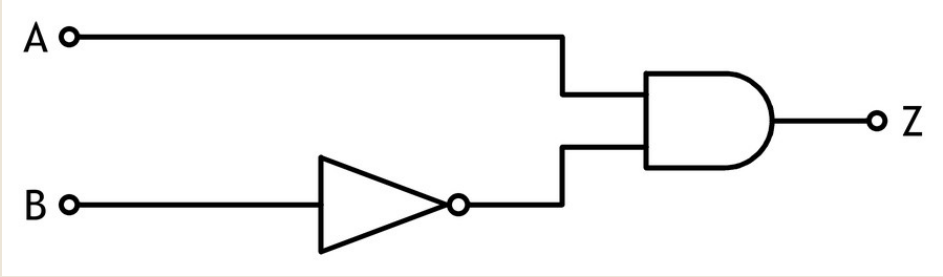
2 **AND**, **OR** or **NOT**?

3 **Write** the Boolean.

4 **Draw** it, **check** it with a truth table.



Worked Example 1: A Fan



A **fan** runs when it is **hot** and the **door** is **not** open.

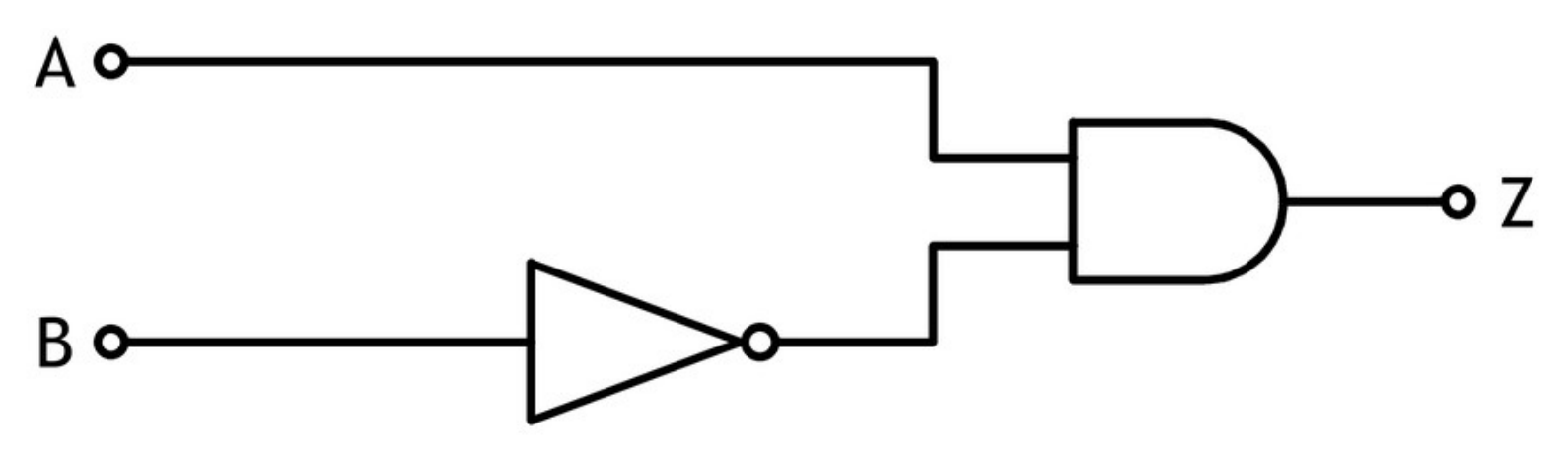
1 A = hot, B = door open. Z = fan.

2 AND, with **NOT** on B.

3 $Z = A \bullet \overline{B}$

4 Draw it. Check it next.

Worked Example 1: Check It



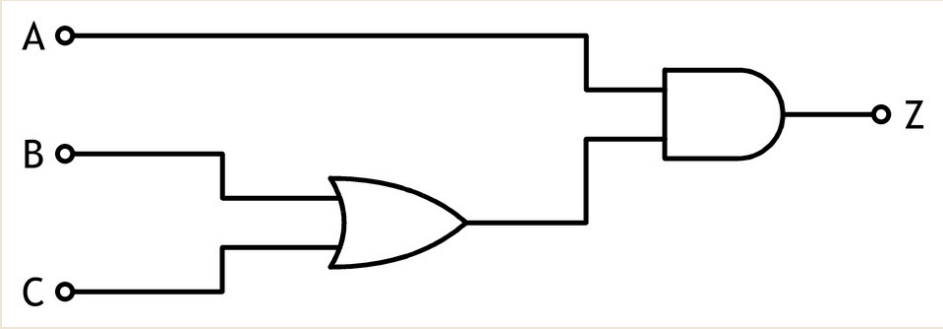
Check: is Z = 1 only when hot and the door is shut?

$D = \overline{B}$

$Z = A \bullet D$

A	B	D	Z
0	0	1	0
0	1	0	0
1	0	1	1
1	1	0	0

Worked Example 2: The Lift



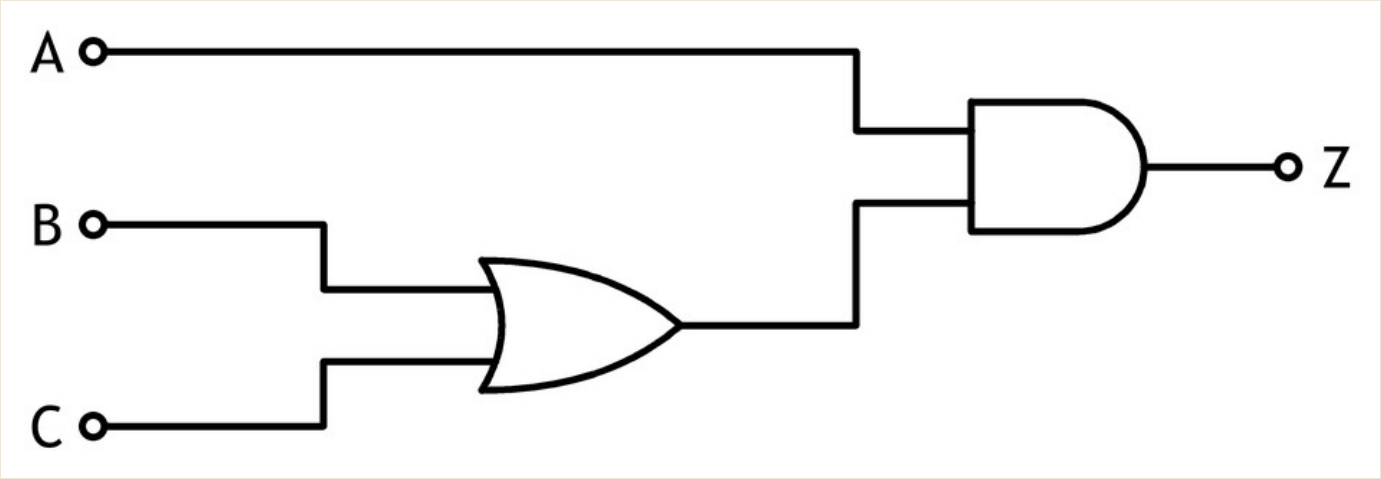
Lift moves when the **door** is closed and the **floor** or **call** button is pressed.

1 A = door, B = floor, C = call.

2 B **OR** C, then **AND** A.

3 **Z = A • (B + C)**

Worked Example 2: Check It



Check: does the lift move only with the door closed?

D = B + C

Z = A • D

A	B	C	D	Z
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

Show Me: Write Z



1. Bell: **either** of two buttons.
2. Kettle: **switch on** and **water in**.
3. Night light: it is **not** light.
4. Gate: **card** and **not** locked.
5. Beep: door **A** or **B** open, and key in (**C**).

$Z = A + B$

$Z = A \cdot B$

$Z = \overline{A}$

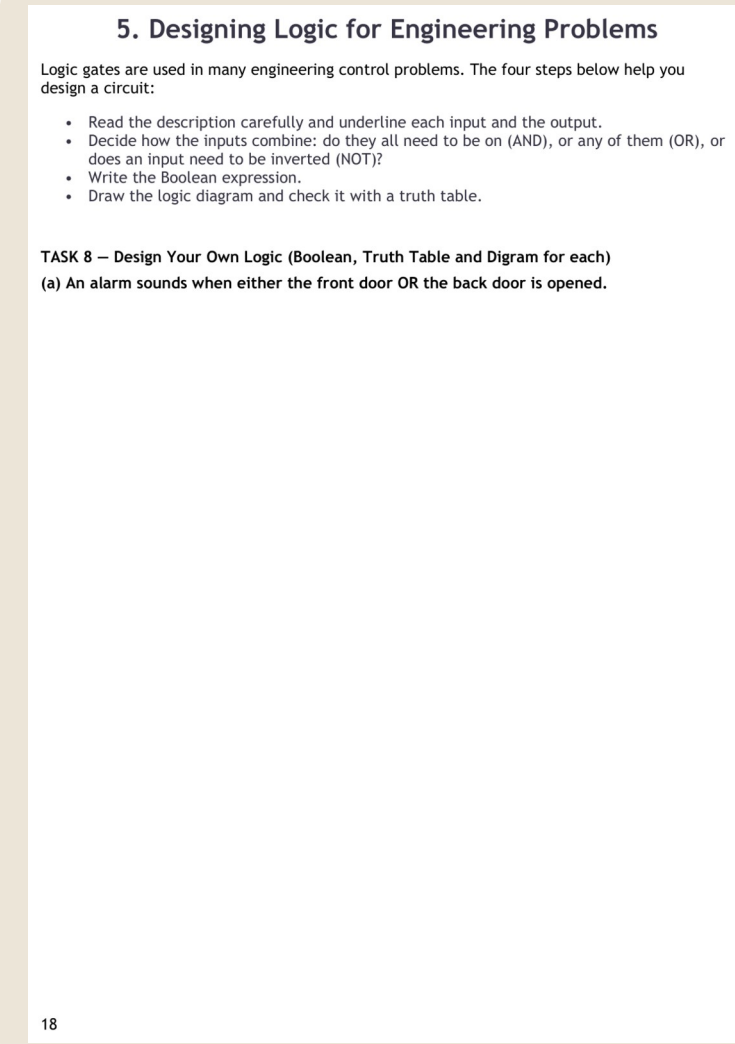
$Z = A \cdot \overline{B}$

$Z = (A + B) \cdot C$

Task 8 (B4a-L5-p18)

Task 8

Write the **Boolean** for each problem.
Complete a **truth table** and **draw** the diagram.
Pages **18** to **20**.



5. Designing Logic for Engineering Problems

Logic gates are used in many engineering control problems. The four steps below help you design a circuit:

- Read the description carefully and underline each input and the output.
- Decide how the inputs combine: do they all need to be on (AND), or any of them (OR), or does an input need to be inverted (NOT)?
- Write the Boolean expression.
- Draw the logic diagram and check it with a truth table.

TASK 8 — Design Your Own Logic (Boolean, Truth Table and Digram for each)

(a) An alarm sounds when either the front door OR the back door is opened.

18

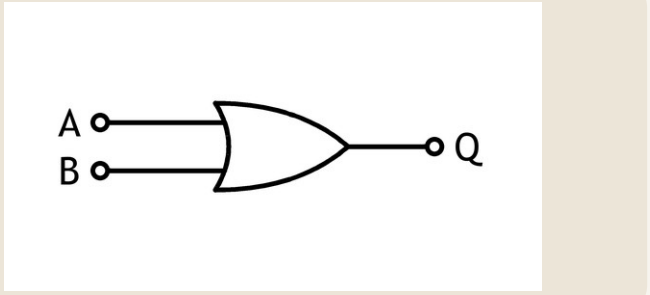
Answers: Task 8 (1 of 2)

Part	Inputs	Boolean
(a)	A front door, B back door	Q = A + B
(b)	A door closed, B start	Q = A • B
(c)	A dark, B movement	Q = A • B
(d)	A stop button (1 = not pressed)	Q = A

Answers: Task 8 (2 of 2)

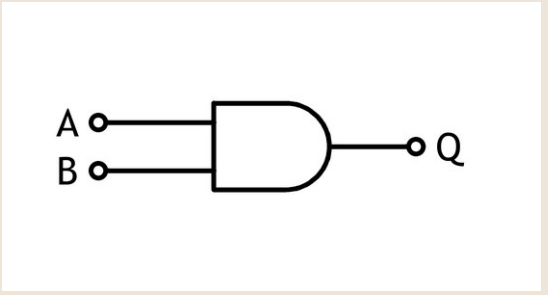
(a) OR

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1



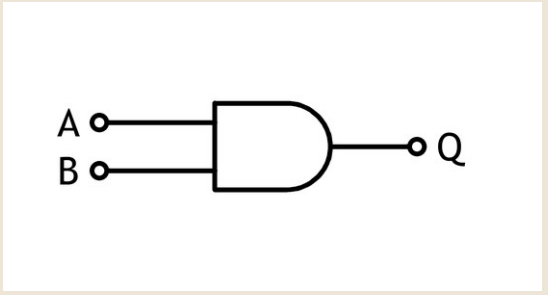
(b) AND

A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1



(c) AND

A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1



Checkpoint



1. **State** the first design step.
2. **Explain** why we check with a truth table.
3. '**Either**' button. **State** the gate.

Underline inputs, output

Every case, **so** no surprises

OR

Look back — by the end of this part…



1. I can **pick out** the **inputs** and the **output**
2. I can **choose** AND, OR or NOT and **write** the Boolean
3. I can **draw** the diagram and **check** it with a truth table

Daily Review



1	Z = A • (B + C). **State** the gate done **first.**	OR
2	'**Both**' switches. **State** the gate.	AND
3	**State** the rows for **3** inputs.	8
4	**State** what a **bar** over B means.	NOT B
5	**State** what Z is for **AND** with 1, 0.	0

PART 2

Build It, Then Work Back

We are learning to build and test a design, and to write Boolean from a truth table.

By the end of this part…



1. I can **build** a design on the **logic board**
2. I can **test** all **8** rows and **compare**
3. I can **write** Boolean from a **truth table**

PRACTICAL

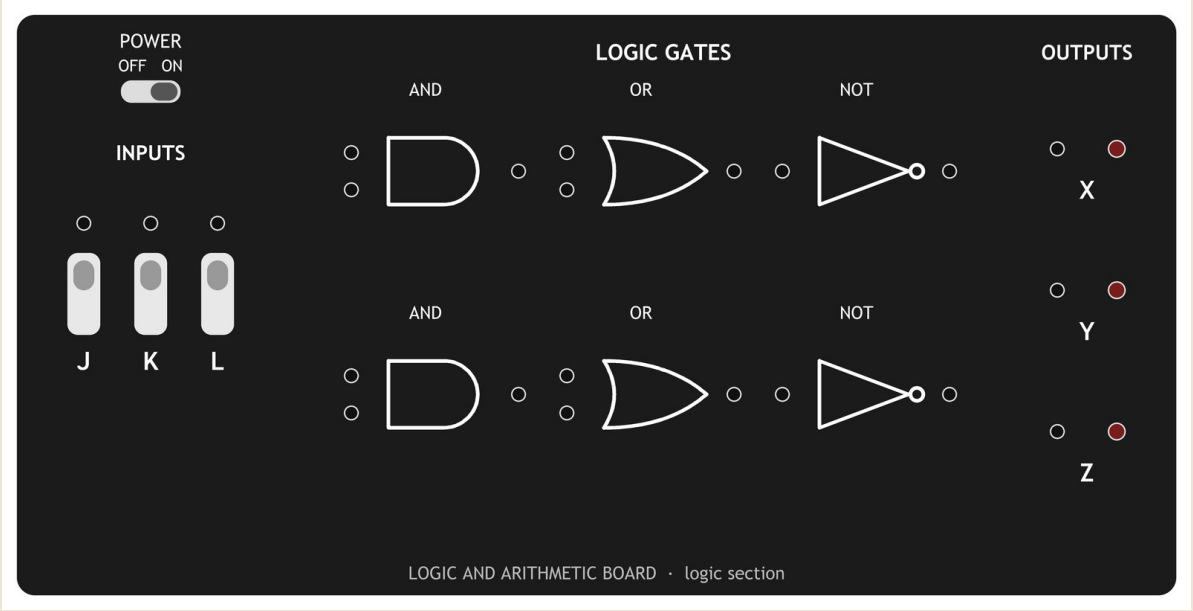
Build the Lift

Build Z = A • (B + C) on the logic board, then test all eight rows.

What to Collect

Collect one set per pair.

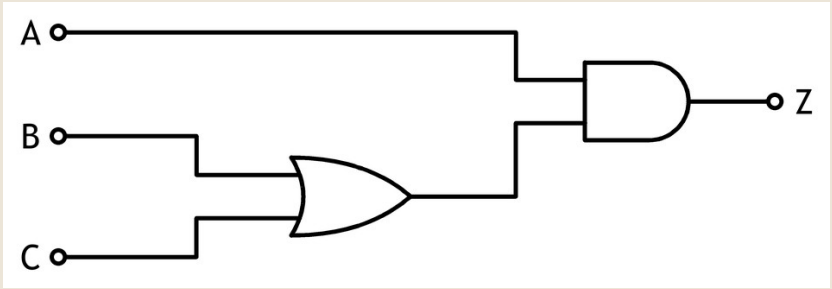
1. **Logic and Arithmetic** board
2. **Flying leads** (5)
3. Your **jotter** for the table
4. Hold the board by its **edges**.



Practical: Build the Lift

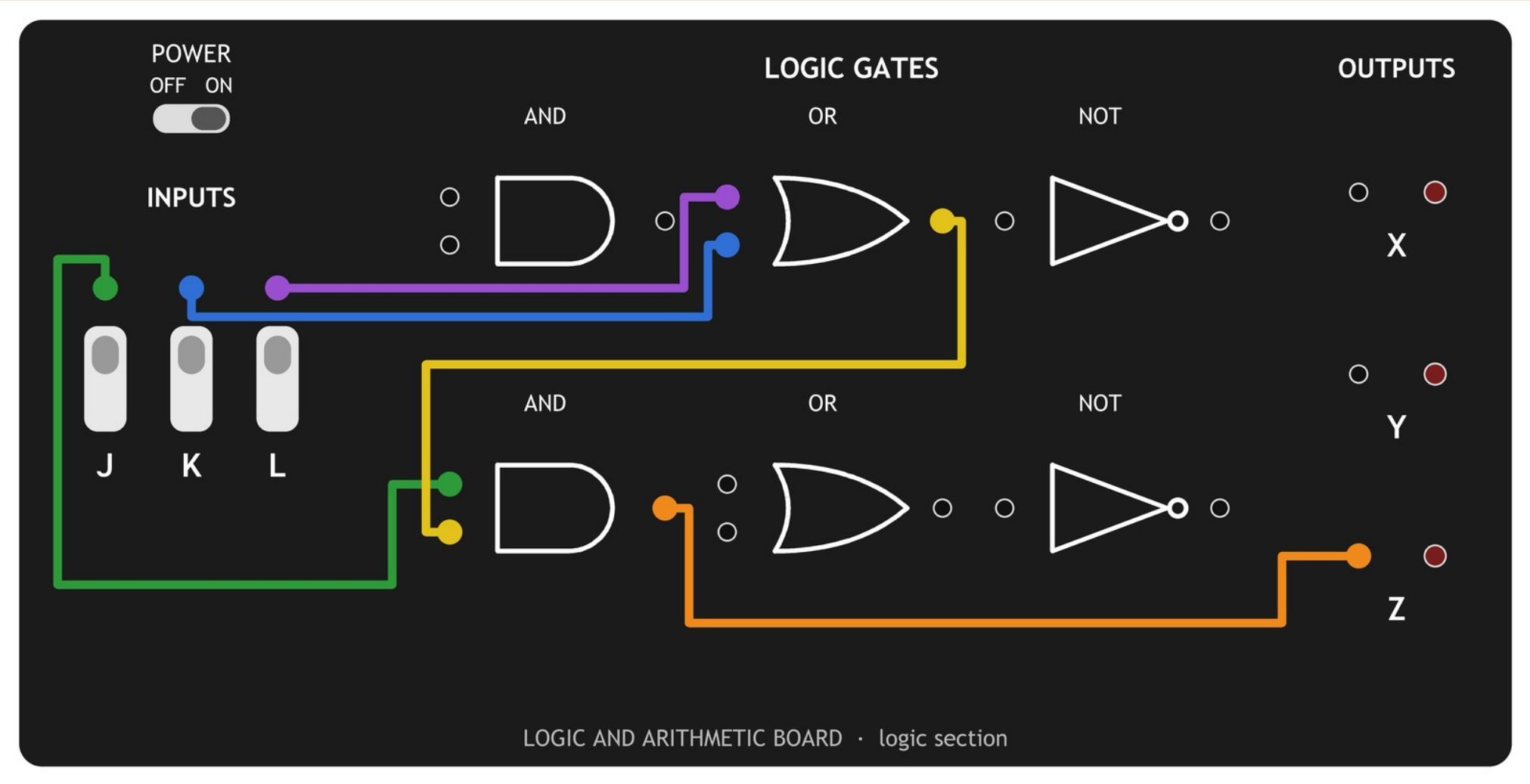
Build It (jotter)

Connect J (A), K (B), L (C) as the bench card.
Test all **8** rows in order. **Record** Z.
Compare with Worked Example 2.



SAFETY Never join two outputs.

Build It: Bench Card 1



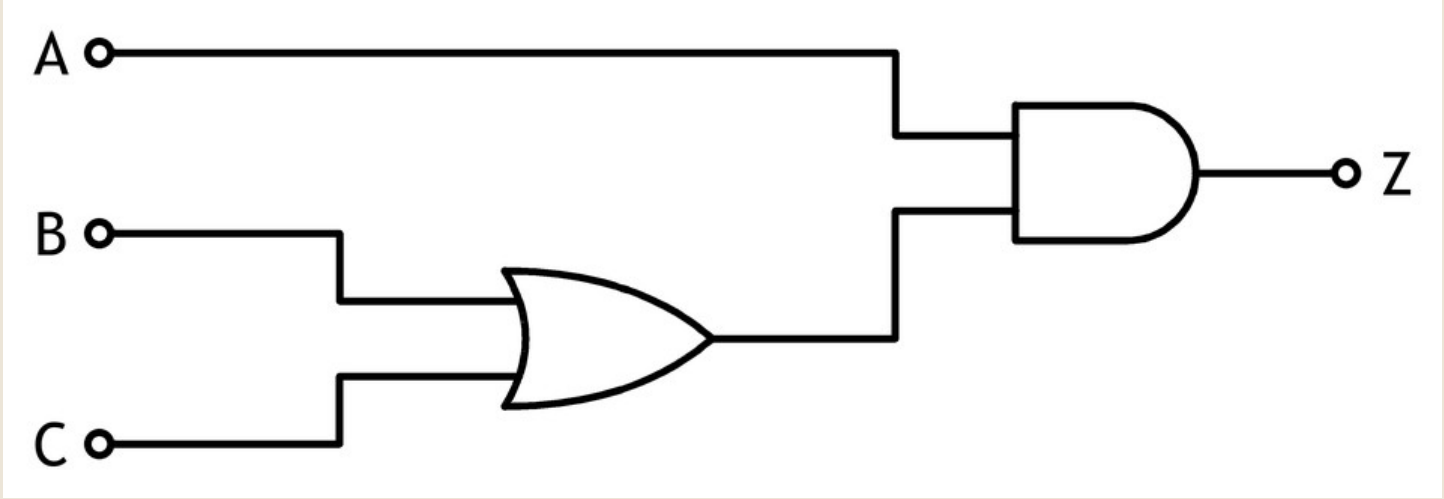
No Kit at Your Bench? Yenka

Three **switches**, labelled **A**, **B**, **C**.

B and C into **OR**.

OR and A into **AND**.

LED to 0 V. Test **8** rows.



Answers: Build the Lift



A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

From a Truth Table to Boolean



1 Find each row with **Z = 1**.

2 AND its inputs: a **bar** on each **0**.

Row 0 1: $(\overline{A} \cdot B)$

Row 1 0: $(A \cdot \overline{B})$

3 **OR** the brackets:

$Z = (\overline{A} \cdot B) + (A \cdot \overline{B})$

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0

2023 Section 2 Question 12 (g)

The truth table for another logic circuit in the washing machine is shown below.

F	G	H	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

(g) Complete the Boolean equation for output Y in terms of inputs F, G and H. **3**

Y = _____

3 marks.

Rows with **Y = 1**:

one **bracket** each,

then **OR** them.

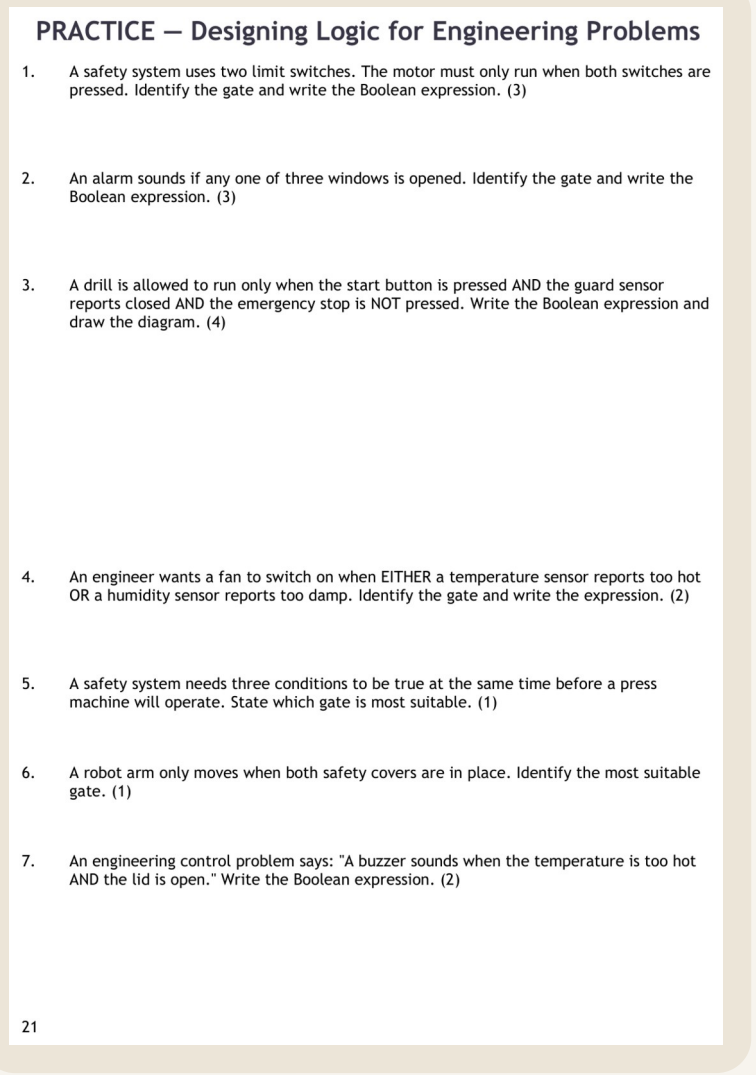
Marking: 2023 Section 2 Question 12 (g)

Question	Expected response	Max mark	Additional guidance
12 (g)	$Y = (F \bullet \overline{G} \bullet H) + (F \bullet G \bullet \overline{H})$	3	1 mark for $(F \bullet \overline{G} \bullet H)$. 1 mark for $(F \bullet G \bullet \overline{H})$. 1 mark for OR-ing all statements.

Review Questions (Booklet pages 21–22)

Review Questions

Answer Questions **1** to **10** on your own.
Questions **8** to **10** are on **page 22**.



PRACTICE — Designing Logic for Engineering Problems

1. A safety system uses two limit switches. The motor must only run when both switches are pressed. Identify the gate and write the Boolean expression. (3)

2. An alarm sounds if any one of three windows is opened. Identify the gate and write the Boolean expression. (3)

3. A drill is allowed to run only when the start button is pressed AND the guard sensor reports closed AND the emergency stop is NOT pressed. Write the Boolean expression and draw the diagram. (4)

4. An engineer wants a fan to switch on when EITHER a temperature sensor reports too hot OR a humidity sensor reports too damp. Identify the gate and write the expression. (2)

5. A safety system needs three conditions to be true at the same time before a press machine will operate. State which gate is most suitable. (1)

6. A robot arm only moves when both safety covers are in place. Identify the most suitable gate. (1)

7. An engineering control problem says: "A buzzer sounds when the temperature is too hot AND the lid is open." Write the Boolean expression. (2)

21

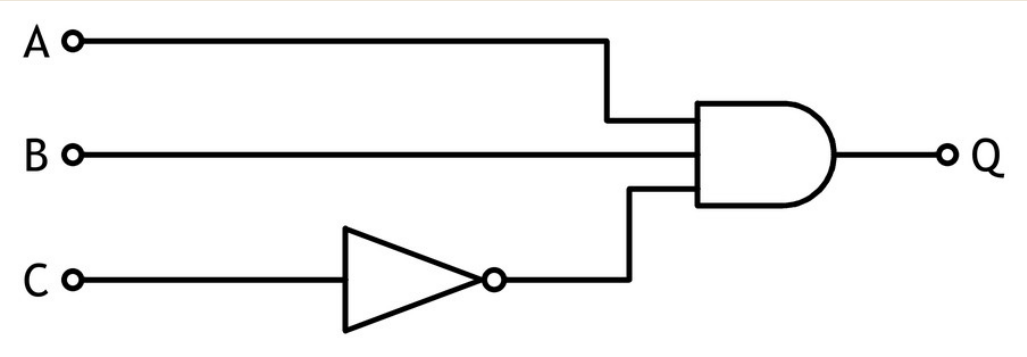
Answers: Review Questions (1 of 3)



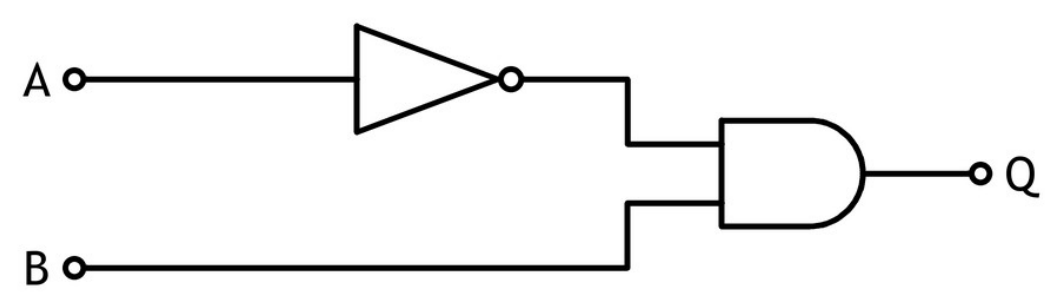
Q	Answer
1	**AND**; Q = A • B
2	**OR**; Q = A + B + C
4	**OR**; Q = A + B
5	**AND** (three inputs)
6	**AND**
7	Q = A • B

Answers: Review Questions (2 of 3)

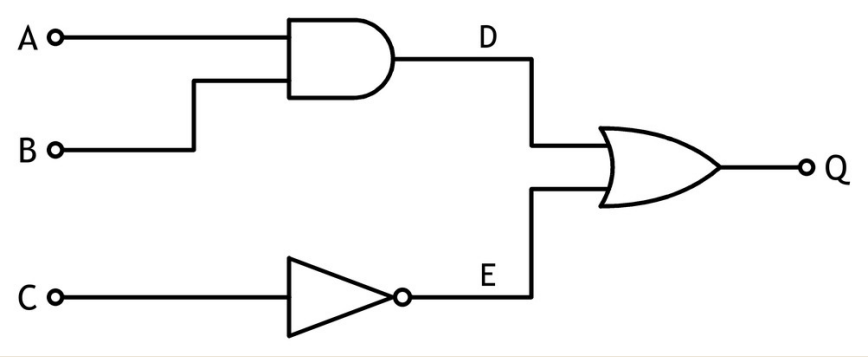
Q3: $Q = A \bullet B \bullet \overline{C}$



Q8: $Q = \overline{A} \bullet B$



Q9: $Q = (A \bullet B) + \overline{C}$



Answers: Review Questions (3 of 3)

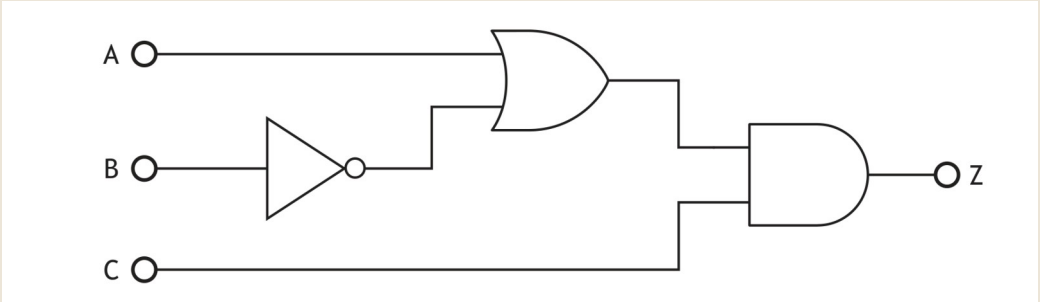


A	B	C	Q
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Logic in the Exam

Year	Questions	Marks
2026	Q5, Q14(d)	6
2025	Q11(c), (d)	6
2024	Q1, Q12(c), Q14(c)	9
2023	Q3, Q12(f), (g)	8
2022	Q12(a), (b)	6

Scan the QR for every past paper by topic.



2026 S1 Q5

Checkpoint

1. Z = 1 for **0 1** and **1 1**. **Write** Z.

2. **Explain** why inputs are **labelled**.

3. **State** what to check if a row is wrong.

$(\overline{A} \bullet B) + (A \bullet B)$

To test, **so** we know which

The leads

Look back — by the end of this part…



1. I can **build** a design on the **logic board**
2. I can **test** all **8** rows and **compare**
3. I can **write** Boolean from a **truth table**

Extension



1. A car alarm sounds when the **alarm is set** and a **door** or the **boot** opens. **Design** it.

2. **Write** Z from a table with Z = 1 only in row **1 1 0**.

Course Specification — 1 of 1

Digital electronic control systems

- AND, OR and NOT gates and combinations with up to three inputs, using truth tables, logic diagrams and Boolean expressions

Credits (1 of 2)

Title: Photo: Fanti Salms, CC BY-SA 3.0, via Wikimedia Commons

Drawings: Gate symbols, logic diagrams and circuits drawn by R Stewart; booklet pages from Booklet 4a, Logic and Digital Control.

Screenshots and page figures: Booklet 4a, Logic and Digital Control: page images, Berwickshire High School.

Logic board: Logic board drawn by R Stewart from the UKESF / University of Southampton Logic and Arithmetic kit teacher guide.

Bench card: Logic board drawn by R Stewart from the UKESF / University of Southampton Logic and Arithmetic kit teacher guide.

2023 Section 2 Question 12 (g): © Qualifications Scotland (SQA). 2023 National 5 Engineering Science Section 2, Q12

Marking: 2023 S2 Q12(g): © Qualifications Scotland (SQA). 2023 National 5 Engineering Science Section 2, Q12(g) marking instructions

Credits (2 of 2)

Logic in the Exam: © Qualifications Scotland (SQA). National 5 Engineering Science 2022-2026 question papers; figure 2026 Section 1, Q5

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R Stewart - November 2026